

CLASSIFICATION OF HAND-WRITTEN DIGITS

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1. LOGISTIC REGRESSION FOR MULTI-CLASS CLASSIFICATION

Logistic regression is a commonly used model for classification tasks. In our case we want to use logistic regression for multi-class classification, which is a generalization of the binary logistic regression.

In our case, we want to predict the class of a given 8×8 grayscale image of a hand written digit. One would expect that a given input can not be classified as more than one digit at the same time. Therefore, we want to compute the probability/score of a given input image to belong to every of the 10 classes (digits from 0 to 9) and assign the class with the highest probability/score to the input image.

1.1. Our problem.

An input image is represented as a vector $a_j \in \mathbb{R}^{64}$. Each entry of the vector is an integer between 0 and 16 and corresponds to the intensity of a pixel in the image with 0 being the lowest intensity and 16 being the highest intensity. Each image a_j in the dataset is associated with a label $y_j \in \{0, \dots, 9\}$, which is the digit that is represented in the image. In order to use the labels in our model, we encode them into one-hot vectors $p_j \in \{0, 1\}^{10}$, where the i -th entry is 1 if the label is i and 0 otherwise. [Dev]

We want to train a function $\varphi : \mathbb{R}^{64} \times (\mathbb{R}^{64} \times \mathbb{R})^{10} \rightarrow \mathbb{R}^{10}$ that takes an input image a_j and a parameter $x = (w_i, b_i)_{i=0}^9$ and outputs a vector of scores for each class. The score for class i is computed as $\varphi(a_j; x)_i = a_j^T w_i + b_i$. The $w_i \in \mathbb{R}^{64}$ is a weight vector and $b_i \in \mathbb{R}$ is a bias term for class i .

The training of the model consists of the following optimization problem

$$\underset{x \in (\mathbb{R}^{64} \times \mathbb{R})^{10}}{\text{minimize}} \sum_{j=1}^N L(p_j, \varphi(a_j; x)),$$

with $N \in \mathbb{N}$ being the number of training samples and L the cross entropy loss defined as

$$(1.1) \quad L(p, s) = \log \left(\sum_{l=0}^9 \exp(s_l) \right) - \sum_{i=0}^9 p_i s_i.$$

After the model has been trained, we can use it to classify new input images a by computing $\varphi(a; x)$ and assigning the class with the highest score to the input image, i.e.

$$\tilde{y} = \arg \max_{i \in \{0, \dots, 9\}} \varphi(a; x)_i.$$

The values of $\varphi(a_j; x)_i$ can be interpreted as probabilities of the input image a_j belonging to class i after applying the softmax function, i.e.

$$\sigma(s)_i = \frac{\exp(s_i)}{\sum_{l=0}^9 \exp(s_l)}.$$

1.2. Solvability.

The given optimization problem is a convex problem. This is because the log-sum-exp function is convex as we have already proved in the lecture. For reference one can also check [Boy09, pp. 72, 74]. Also the second term of our loss function (1.1) is linear and therefore convex. Also, φ is an affine mapping in x . And at the same time convex function over an affine function is also convex (cf. [Boy09, p. 67]).

In general however, the optimization problem is not necessarily solvable. This is because the loss function (1.1) is not strictly convex in its scores s and therefore not strictly convex in the parameter x .

This, is because the loss function (1.1) is invariant under the addition of the same constant to every score s_i . More precisely, for a constant $c \in \mathbb{R}$ we have

$$\log \left(\sum_{l=0}^9 \exp(s_l + c) \right) = \log \left(\exp(c) \sum_{l=0}^9 \exp(s_l) \right) = \log \left(\sum_{l=0}^9 \exp(s_l) \right) + c$$

and also

$$\sum_{i=0}^9 p_i(s_i + c) = \sum_{i=0}^9 p_i s_i + c.$$

Therefore, we have

$$L(p, s + c\mathbb{1}) = \log \left(\sum_{l=0}^9 \exp(s_l + c) \right) - \sum_{i=0}^9 p_i(s_i + c) = L(p, s) + c - c = L(p, s).$$

This implies that for a solution x of our optimization problem, we get arbitrary additional solutions, e.g. $x' = (w_i + c\mathbb{1}, b_i)_{i=0}^9$ for any constant $c \in \mathbb{R}$. Therefore, the optimization problem is not uniquely solvable.

Even if we ignore non-uniqueness, the minimum in general may fail to be attained at all. A reason is linear separability of the training data. For

$$s_{j,i} := \varphi(a_j; x)_i = a_j^\top w_i + b_i,$$

and $p_{j,y_j} = 1$, we can rewrite the loss as

$$(1.2) \quad L(p_j, s_j) = \log \left(\sum_{l=0}^9 e^{s_{j,l}} \right) - s_{j,y_j} = \log \left(1 + \sum_{i \neq y_j} e^{-(s_{j,y_j} - s_{j,i})} \right) \geq 0.$$

Hence $L(p_j, s_j)$ gets small when all the $s_{j,y_j} - s_{j,i}$ are large, i.e. when the correct class score s_{j,y_j} dominates all other scores $s_{j,i}$ by a large margin $\gamma > 0$. If this is the case, we call the dataset *separable*. We can then scale the parameters by any factor $t > 0$, i.e. we consider

$$x^{(t)} := (tw_i, tb_i)_{i=0}^9.$$

Then $s_{j,i}^{(t)} = \varphi(a_j; x^{(t)})_i = t s_{j,i}$ and the differences scale as

$$s_{j,y_j}^{(t)} - s_{j,i}^{(t)} = t(s_{j,y_j} - s_{j,i}) \geq t\gamma.$$

If we use this, the above expression (1.2) yields

$$0 \leq L(p_j, \varphi(a_j; x^{(t)})) = \log \left(1 + \sum_{i \neq y_j} e^{-t(s_{j,y_j} - s_{j,i})} \right) \leq \log(1 + 9e^{-t\gamma}) \xrightarrow{t \rightarrow \infty} 0.$$

Therefore, the optimization term $\sum_{j=1}^N L(p_j, \varphi(a_j; x^{(t)}))$ gets arbitrarily close to 0, but there is no minimizer to the problem.

Remark 1.2.1. The idea, that linear separability can be a problem for logistic regression, was inspired by [Ada18, Linear Separability and Regularization].

1.3. Regularization.

solves some problems **explain why regularization is a good thing to do!**

2. STOCHASTIC GRADIENT DESCENT

In order to solve the optimization problem of our logistic regression model, we use stochastic gradient descent.

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